

Coins 326 Ethics Paper

Grade Received: A

This ethics paper was an essay commenting on the struggle between the desire for free speech and the need for a just society to protect its citizens from undo harassment.

Coins 431 - Advanced Operating Systems Presentation II

Grade Received: A

This presentation was delivered on the new Windows 7 Virtual PC feature. This presentation charts the history of Microsoft virtualization technology, and includes a few performance evaluations of Virtual PC performance. The performance of Windows 7 Virtual PC is compared to a popular virtualization package VMWare's VMPlayer.

Note, the performance of the virtualized 2D graphics in VMPlayer vs. the native Windows 7 graphics. This appears to be a video driver defect.

Coins 431 - Advanced Operating Systems Ethics Paper

Grade Received: B+

This ethics paper is an essay on the effects of video game violence. I took a perhaps controversial approach to advocate violence in video games as a way to offset what I see as the pacification of American society.

Violence in Video Games:

Video Game Violence Can Offset Passive Culture

David Coons

COIN 431

Gary Underwood

September 10th, 2009

American society has long obsessed with protecting its citizens from harmful influences. From the first puritan settlements in New England, through the Parents Music Resource Center in the mid 1980's, to the Janet Jackson FCC scandal, America has always tried to protect its population from harmful material. One of the latest areas of concern is the level of violence in video games. When video games were first introduced, video games were seen as quite harmless. With the exponential increase in realism that has accompanied by the exponential increase in computing power, however, the issue has warranted increased scrutiny.

The 1980's, arguably the hey-day of the video arcade, was the decade that the world fell in love with video games. The games, though popular and addicting, were very simple and cartoonish. Though there was violence, it was limited to shooting bugs in Galaga or zapping tanks in Combat. That all changed, however, in the early 1990's. In 1991, street fighter was released. Then, shortly thereafter, Midway released its own street fighter type game with over-the-top violence, Mortal Kombat, and everything changed (Fahs, 2008). Mortal Kombat used digitized live actors instead of hand-drawn figures for increased realism. The game also featured gratuitous blood and gore, most notably in the 'fatalities' which included such acts as pulling a still beating heart from the victim's chest, or ripping the spine out of a living victim (Figure 1).



Figure 1 : Mortal Kombat Fatality (Lexia, 2005)

This dramatic increase in video game violence led to US congressional hearings on video game violence. Congress offered the video game industry the same offer it made to the record industry a few years prior; institute a ratings system yourself, or Congress will legislate one. Hence, after

a few false starts, was born the Entertainment Software Ratings Board, or ESRB (Chalk, 2007). The ESRB rates video games on several various factors ranging from 'Alcohol Reference' to 'Sexual Content' to 'Simulated Gambling'. The ESRB then assigns one of the following ratings: Early Childhood, Everyone, Everyone 10+, Teen, Mature, or Adults Only (Game Ratings & Descriptor Guide, 2009).

The obvious question that is still unanswered, however, is: Are violent video games bad?

In order to answer the above question, you first must define what "bad" is. This is where things move out of the realm of hard science and into more nebulous concepts of social engineering. First, you must define what effects violent video games have on people, not an easy task. Then, you must ascertain whether these effects move society closer or further away from the ideal. This of course begs the question, what is the ideal society?

While the jury may still be out on the precise effects of video game violence, the general consensus is that they promote aggression. Studies have found that exposure to violent video games increases aggression (Harding, 2009). Of course, violent video games also appeal to aggressive people. Eric Harris and Dylan Klebold were rabid fans of the ID Software's controversial video game Doom, which was one of the first popular 3d first-person "shoot-em-up" games. A critic of video game violence could make the argument that the violent video game was partially responsible for the tragedy of Columbine. However, in my opinion, the

tragedy of Columbine was made worse by the LACK of aggression on the part of the victims and police.

Eric Harris and Dylan Klebold were bullies, and bullies only prey on the weak. One of the biggest problems that face our society is what many have termed the "Feminization of America". Prior to the industrial revolution, boys grew up alongside their fathers. Most were farmers who grew up wringing a meager existence from the land by the sweat of their own brow. With the rise of modern industry, men went off to work and boys were raised by their mothers and by school teachers, usually female. Boys found it difficult to learn to be men (Farrar, 2009). The second blow came with the feminist movements of the 20th century. Modern feminism teaches that male tendencies such as aggression, domination, and independence are wrong (Farrar, 2009). One of the goals of the political correctness movement is to punish exceptionalism and encourage mediocrity. Many schools have removed dodge ball from the PE class because it is 'traumatic' to the weaker students (Reilly, 2001).

Movie heroes, up until the 1960's, were very masculine. John Wayne is a classic example. He was gruff, tuff, and independent. Westerns were very popular, and they romanticized tough men in a lawless land. Then, around the 1960's, something happened. Baby boomers and their leaders reacted



Figure 2: Flower Power, taken by Bernie Boston at an anti-war protest in Washington on Oct. 22, 1967 (Valbones, 2008)

against the masculine society of the 1950's and began to preach more feminine qualities like compassion, peace, and compromise. The famous picture of the hippie pushing a flower into the barrel of a soldier typifies this philosophy (Figure 2). Many view this as the conquering of the military by peace. Of course, if the soldier had pulled the trigger, the "power" of the "flower" would have been shown to be quite lacking.

There are still masculine role models of course. The governor of California has made quite a living embracing and displaying masculinity. However, for every "Terminator", there seems to be a dozen "Outbreak's". Outbreak was a film released in 1995 and featured Dustin Hoffman as an Army doctor who fights a viral outbreak that was unwittingly released from a covert US bio-weapons lab. The climactic moment is when Dustin Hoffman faces off in against an Air Force bomber ready to bomb the infected town. Dustin Hoffman saves the day by appealing to the pilot's intellect and emotions, not by just shooting down the plane with the missiles he possesses. After that, the 'good' general arrests the 'evil' general because he, too, has been swayed by the

reason and emotional appeal of Dustin Hoffman. The film grossed over \$67 million dollars in 1995, equivalent to over \$96 million in today's dollars.

What if, at Columbine, there had been more Arnold Swarchenegger and less Dustin Hoffmans? What if the police had raced in to take charge, and not waited for more backup? These were grown men with access to body armor and assault weapons afraid to go up against two teenagers with pistols and shotguns (Dutch, 2008). What if, during the 911 attacks, the passengers had been more aggressive? Fifteen men with box cutters were able to control three airplanes and hundreds of passengers and crew because the crew was trained to be passive and cooperative.

Video game violence has been shown to increase aggression. While unchecked aggression can obviously be a problem, aggression is by itself is in fact a good thing. Aggression is the force that promotes change and action. What this society needs are more people who are more aggressive, not less. David DeAngelo, proprietor of the "Double Your Dating" series, has created a very successful business by teaching men that success with women lies in not being a "wussie" (DeAngelo, 2009). The reason he is so successful is because society has taught men not to be masculine, and we therefore have legions of men who are unsuccessful, un-attractive, lonely, and scared. The cliché of "nice guys finish last" is well known, yet we continue to teach men to be 'nice'. As society grows more and more 'feminine', we surrender power to those who would willingly dominate the weak, be they in government, playgrounds, a high school in Jefferson County, Colorado, or an airplane over New York City.

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Coins 431 - Advanced Operating Systems Presentation I

Grade Received: A

This presentation was delivered on the new Vista memory management features Superfetch and Readyboost. This presentation analyzes the performance benefits of these new technologies, and concludes with some test data.

SUPERFETCH & READYBOOST

Windows Vista Memory Management

Dave Coons

Vista's Improvements

- ▣ Windows Vista was released to the public on January 30th, 2007.
- ▣ Some one of its many new features were Superfetch and ReadyBoost
 - Improvement on virtual memory management
 - Designed to improve responsiveness
 - ▣ Different from performance
 - Responsiveness is how long it takes to start something
 - Performance is how long it takes to finish

Vista Memory Management

- ▣ Vista introduced two new memory management features
 - Superfetch: For machines with plenty of memory
 - Readyboost: For machines with minimal memory

Superfetch

- ▣ An improvement over Windows XP prefetch
- ▣ Monitors computer activity and retrieves page usage histories
- ▣ Directs the Memory Manager to preload data and code from files on disk or from a paging file.
 - Looks for patterns and attempts to preload pages that might be needed later.

Superfetch

- ▣ Essentially extends page-tracking to data and code that was once in memory, but that the Memory Manager has reused to make room for new data and code
- ▣ It stores this information in scenario files with a .db extension in the %SystemRoot%\Prefetch directory alongside standard prefetch files used to optimize application launch
- ▣ Using this deep knowledge of memory usage, SuperFetch can preload data and code when physical memory becomes available

1:<http://technet.microsoft.com/en-us/magazine/2007.03.vistakernel.aspx>

Superfetch

- ▣ Whenever memory becomes, SuperFetch asks the Memory Manager to fetch data and code that was recently evicted
- ▣ This is done at a rate of a few pages per second with Very Low priority I/Os
 - This way the preloading does not impact the user or other active applications

- ▣ 1:<http://technet.microsoft.com/en-us/magazine/2007.03.vistakernel.aspx>

Readyboost

- ▣ Vista's minimum memory requirement is 512k
- ▣ A bit like a fat girl in a tube dress, It will work, but it ain't pretty
- ▣ Readyboost can help improve performance by using flash memory as another option for the backing store

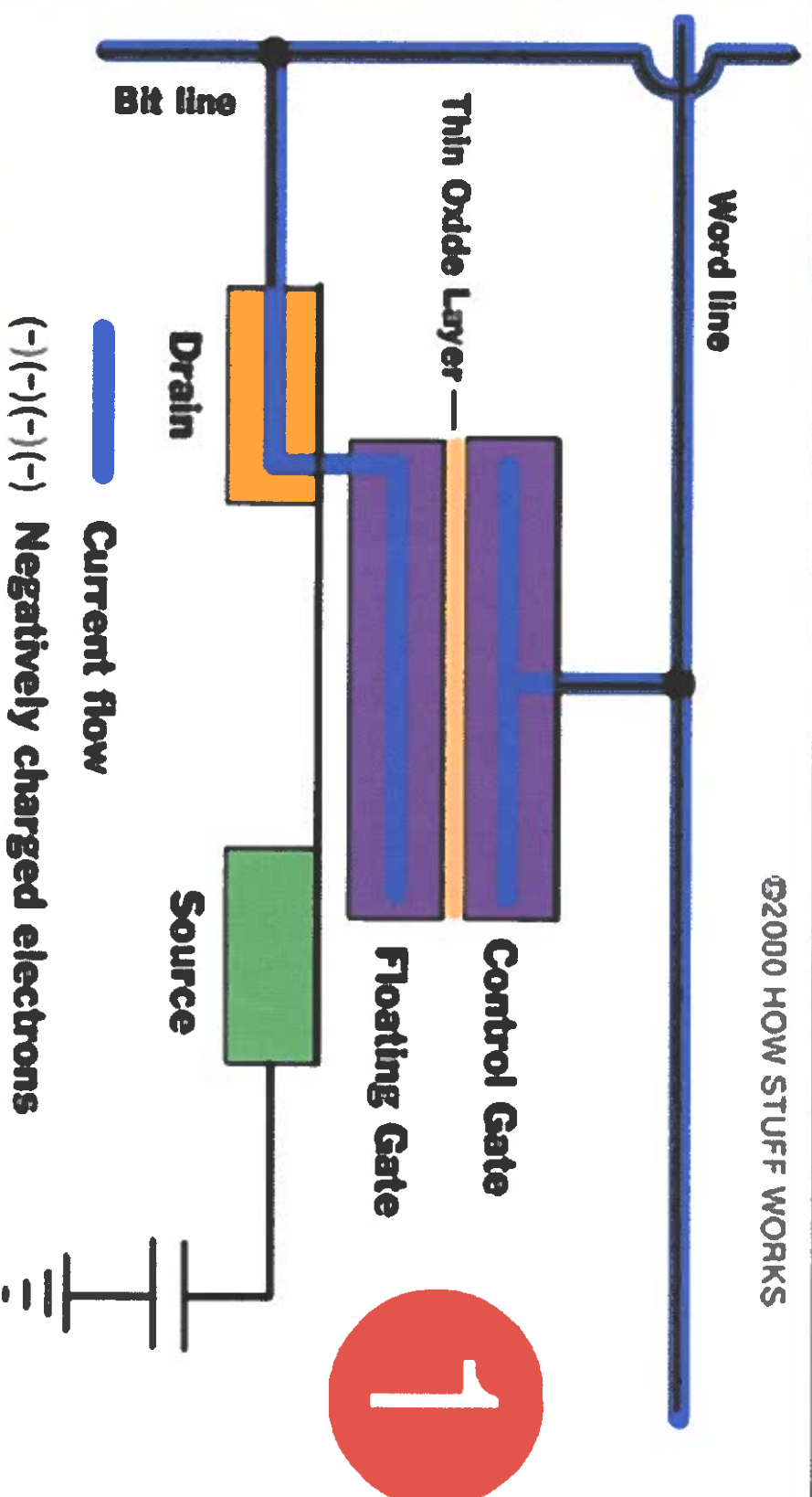


Brief History of ROM Types

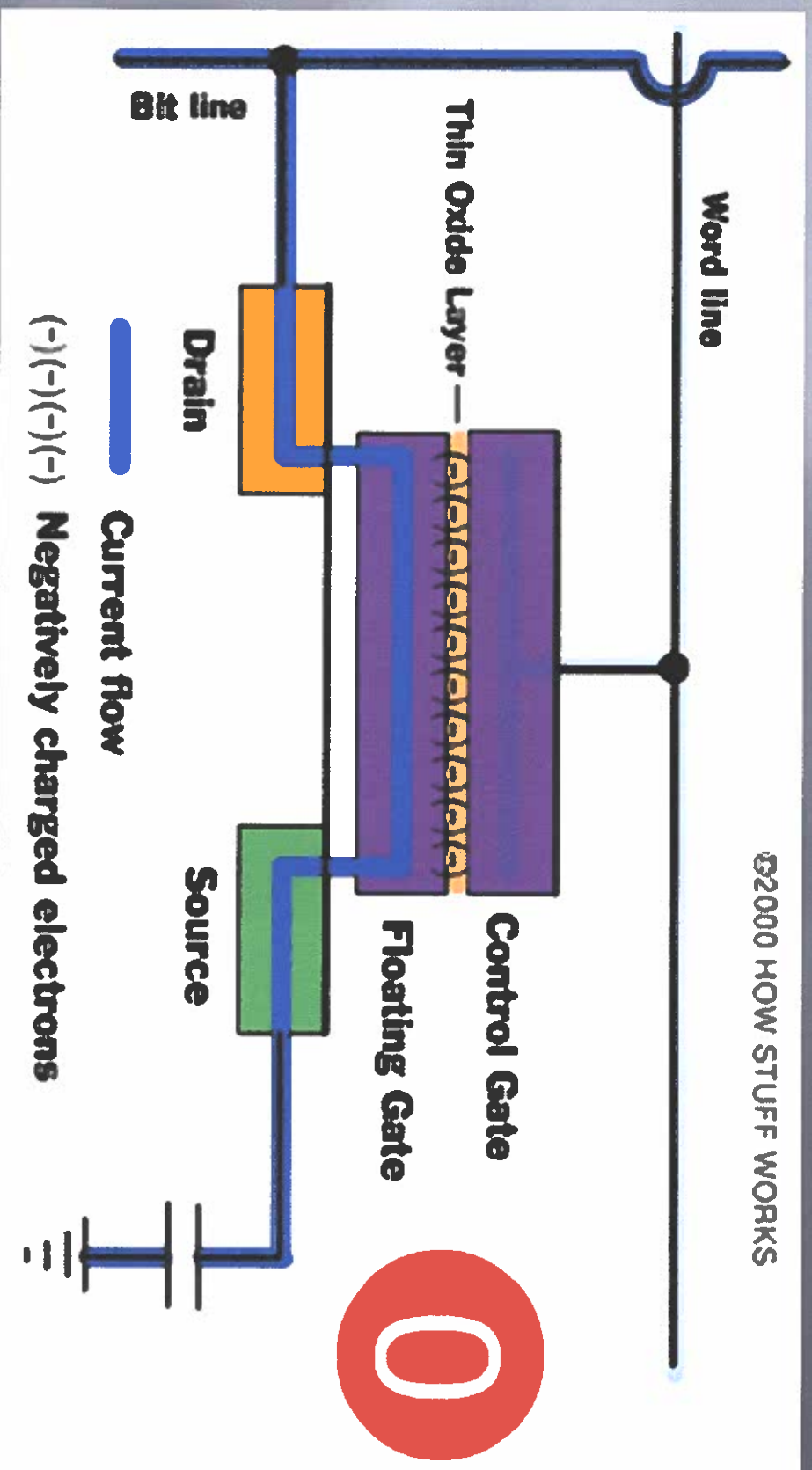
- ▣ ROM: Read Only Memory
 - Like a CD ROM
- ▣ PROM: Programmable Read Only Memory
 - Like a Writable CD ROM
- ▣ EPROM: Erasable Programmable Read Only memory
- ▣ EEPROM: Electronically Erasable Programmable Read Only Memory
- ▣ Flash Memory: Sorta like a cross between PROM and EEPROM

Flash Memory - I

Q2000 HOW STUFF WORKS



Flash Memory - 0



Performance

- ▣ Random Access Times
 - Good USB Drive < 1ms
 - Hard Drive ~ 8ms
- ▣ Read Times
 - Good USB Drive ~ 20 MB/s
 - Hard Drive ~ 60 MS/s
- ▣ Write Times
 - Good USB Drive ~ 14 MB/s
 - Hard Drive ~ 60 MB/s

Performance

- ▣ USB Thumb Drives have slower transfer rates than hard drives, but have MUCH faster seek times
- ▣ Windows Vista and Windows 7 can use thumb drives in the cache hierarchy.
- ▣ Most beneficial when memory is low, and storing small frequently accessed pages, such as the page file

How It Works

- ▣ When a suitable USB drive is detected, the user is asked if they want to use ReadyBoost and how much storage to dedicate to it.
- ▣ Ecache.sys device driver intercepts all read/writes and copies them to the cache file on the thumb drive
- ▣ ReadyBoost compresses and encrypts the data for speed and safety
 - Advanced Encryption Standard (AES) encryption with a randomly generated per-boot session key
- ▣ When ReadyBoost sees random reads that the cache can satisfy, it utilizes it. Sequential reads are passed to the hard drive

Performance Results

▣ Tests...

- First of all we installed Windows Vista Enterprise from scratch and added all required drivers. As we finished the installation we set a system restore point, because we recreated this clean system state (with no impact of SuperFetch learning from our testing activity) for every complete set of benchmarks. We used Microsoft Outlook with a 1.5 GB PST file, the Internet Explorer 7 with a locally stored website and the OpenOffice Writer with a huge press release document as our applications to test SuperFetch and ReadyBoost. Here is our testing sequence:

Source: <http://www.tomshardware.com/reviews/windows-vista-superfetch-and-readyboostanalyzed,1532-5.html>

Performance Results continued...

- ▣ Boot Windows Vista
- ▣ Wait 2 minutes

▣ First Run

- Start Internet Explorer : Wait 5 min (15 minutes if ReadyBoost USB 2.0 Flash device is attached for the first time) : Start Microsoft Outlook : Wait 2 minutes : Start OpenOffice Writer : Wait 2 minutes : Close OpenOffice Writer : Close Microsoft Outlook : Close Internet Explorer : Wait 2 minutes

▣ Second Run

- Start Internet Explorer : Wait 2 minutes : Start Microsoft Outlook : Wait 2 minutes : Start OpenOffice Writer : Wait 2 minutes : Close OpenOffice Writer : Close Microsoft Outlook : Close Internet Explorer : Wait 2 minutes

▣ Third Run

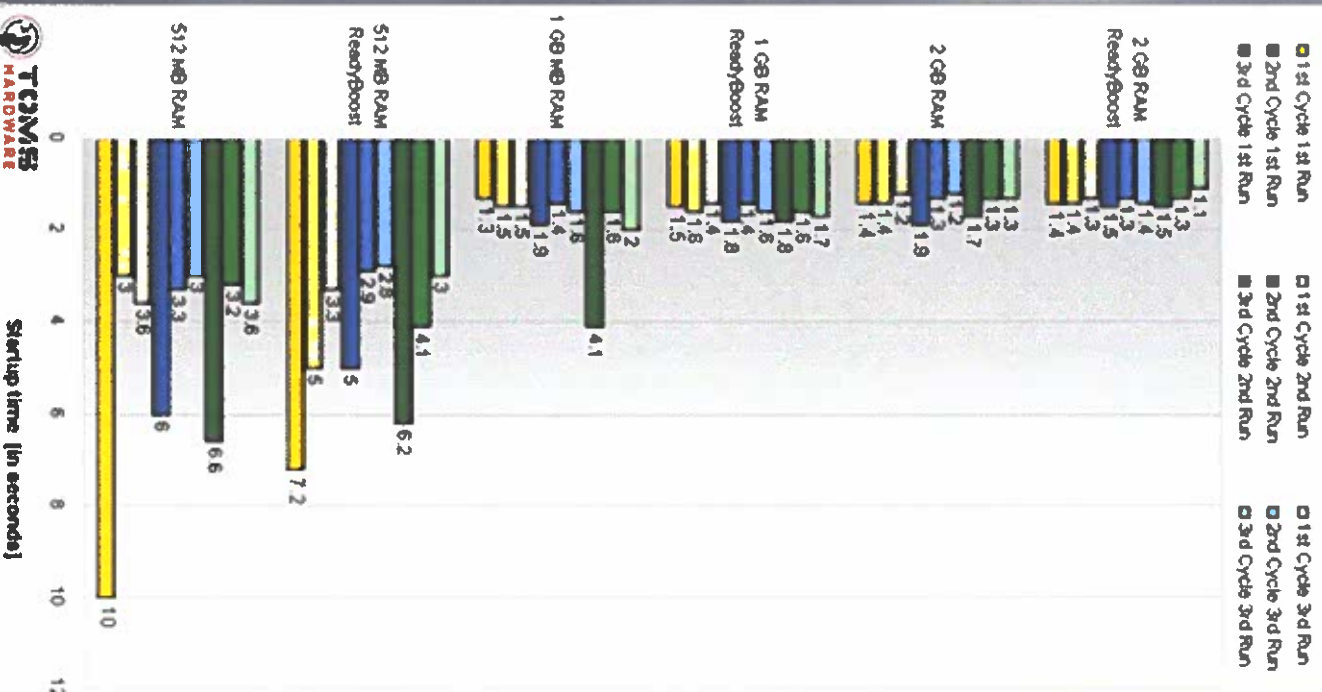
- Start Internet Explorer : Wait 2 minutes : Start Microsoft Outlook : Wait 2 minutes : Start OpenOffice Writer : Wait 2 minutes : Close OpenOffice Writer : Close Microsoft Outlook : Close Internet Explorer :

- ▣ Reboot and start 2nd or 3rd cycle

- ▣ We measured the startup time for Microsoft Office 2007 and the OpenOffice Writer 2.1

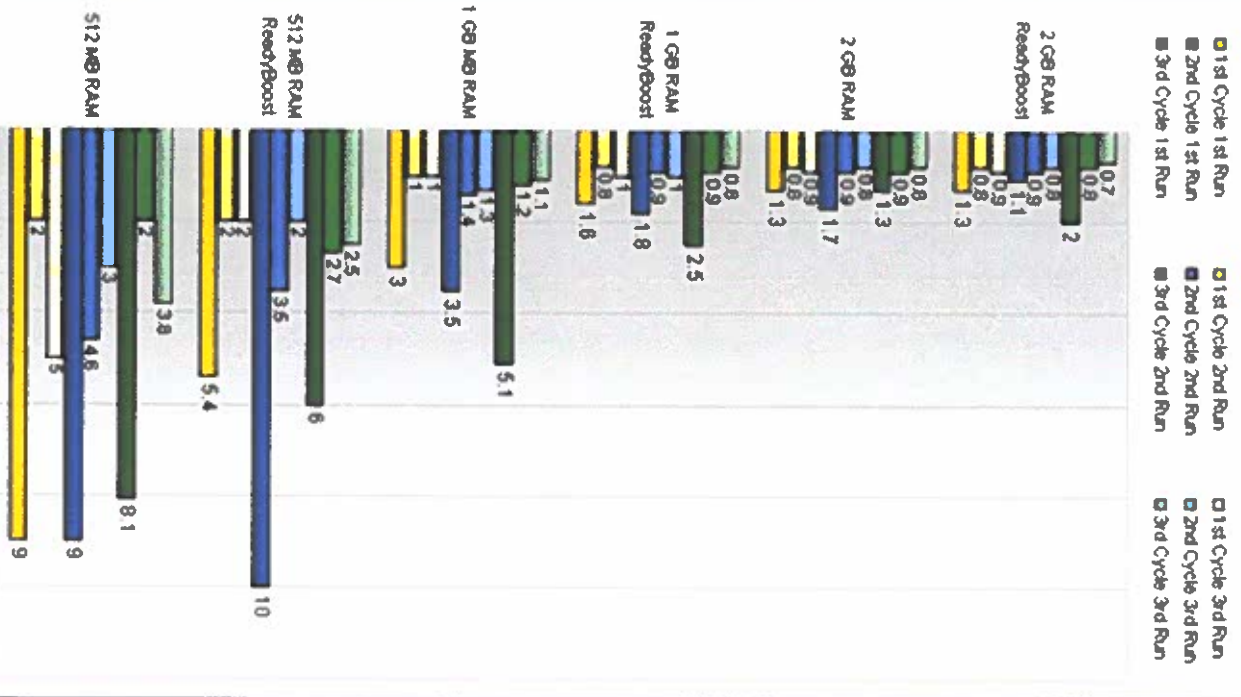
Performance Results continued...

Microsoft Outlook 2007 Start-Up



Performance Results continued...

Open Office Writer 2.1



Conclusions

- ▣ ReadyBoost and SuperFetch can offer significant performance
- ▣ ReadyBoost can dramatically improve performance on low memory systems
 - Still some small gains after memory > 1gb
 - Recommendation: If you have a fast USB thumb drive, might as well use it.
- ▣ SuperFetch does improve responsiveness if it can predict your behavior

Coins 431 - Advanced Operating Systems Presentation II

Grade Received: A

This presentation was delivered on the new Windows 7 Virtual PC feature. This presentation charts the history of Microsoft virtualization technology, and includes a few performance evaluations of Virtual PC performance. The performance of Windows 7 Virtual PC is compared to a popular virtualization package VMWare's VMPlayer.

Note, the performance of the virtualized 2D graphics in VMPlayer vs. the native Windows 7 graphics. This appears to be a video driver defect.

WINDOWS VIRTUAL PC

Dave Coons

COIN 431

Virtual PC History

- ▣ Microsoft's line of PC Virtualization software
- ▣ Originally developed by Connectix
- ▣ Purchased by Microsoft and developed into Microsoft Virtual PC
- ▣ Released as a free product in 2006
- ▣ Upgraded in 2007 as Virtual PC 2007

Source: http://blogs.msdn.com/virtual_pc_guy/archive/2006/07/12/662535.aspx
<http://www.microsoft.com/presspass/press/2003/feb03/02-19PartitionPR.mspx>

Windows Virtual PC

- ▣ More stringent requirements
- ▣ Unlike Microsoft Virtual PC, Windows Virtual PC requires hardware virtualization support
- ▣ Unlike Microsoft Virtual PC, Windows Virtual PC only runs on Windows 7
 - Microsoft Virtual PC 2007 'requires' Windows Vista Business, Windows Vista Enterprise, Windows Vista Ultimate, Windows XP Professional, Windows XP Tablet PC Edition, Windows Vista Service Pack 1 (SP1) (Enterprise, Business, Ultimate), and Windows XP Service Pack 3 (SP3).
 - Microsoft Virtual PC 2007 officially does not support home editions of XP and Vista, but the software will still run on those systems.

Source: <http://www.microsoft.com/windows/virtual-pc/support/virtual-pc-2007.aspx>

<http://www.microsoft.com/windows/virtual-pc/support/requirements.aspx>

Added Features of Windows Virtual PC

- ▣ USB Support
- ▣ Seamless application publishing and launching
- ▣ Support for multi threads
 - Users will be able to run multiple virtual machines concurrently, each running in its own thread, which improves stability and performance.
- ▣ Clipboard sharing
- ▣ Printer redirection
 - Ability to redirect printing from guest machine to printers attached on host machine.

Added Features of Windows Virtual PC

- ▣ Smart card redirection
- ▣ Drive sharing
- ▣ Integration with Windows Explorer
- ▣ Known folder integration between host and guest
- ▣ Support for higher resolutions
 - The allowable resolution for virtual machines has been extended up to 2048x1920 from 1600x1200.
- ▣ Integration with Windows XP Mode

Windows XP Mode

- ▣ Another free package from Microsoft
- ▣ Special Virtual Machine for Windows Virtual PC
 - Includes a free licensed copy of Windows XP Professional
- ▣ Seamlessly integrates with Windows 7
 - Allows software installed on Windows Virtual PC to be launched directly from Windows 7 Start Menu
- ▣ Implemented with Remote Desktop Protocol

Performance

- ▣ Running Passmark's Performance Test 7.0
- ▣ Compare Windows Virtual PC with VMware Player 3.0.0 build-203739
- ▣ Tests run on a Asus G50VT-X1
 - Host OS Windows 7 64 Enterprise Edition
 - 4Gb Ram
 - Intel P8400 @ 2.26GHz
 - nVidia GeForce 9800m GS w/ 512Gb 800MHz DDR3

VMware

- ▣ Principle competition to Microsoft's virtualization technology
- ▣ VMware player also offered free
- ▣ Also simulates a 'standard' pc
 - No exotic video or audio hardware
 - Aim is to have a virtual PC that can run on a variety of host machines, so the LCD is desired

Conclusion

- ▣ Windows Virtual PC has better integration than past incarnations of Virtual PC
- ▣ Windows Virtual PC is slower and less configurable than its competition
- ▣ Windows XP mode is an effective solution for businesses that wish to run legacy software not compatible with Windows 7
 - Performance hit is large, but legacy business software probably does not require modern performance levels
 - No chance of running legacy games